

Correctness Evidence for I/O Optimizations in VPIC-Kokkos: Preliminary Results

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Abstract—Scientific input/output (I/O) optimizations require clear correctness evidence that application developers can review. We study deterministic export tests and scientist-vetted end-to-end analyses within the Kokkos implementation of the Vector Particle-in-Cell (VPIC) application. These validations rely on different oracles and target different properties. This work in progress explores what kinds of correctness evidence should be provided alongside an application I/O optimization.

I. MOTIVATION

Scientific I/O optimizations reduce the cost of writing simulation data while preserving its scientific meaning. But performance gains do not ensure correctness: an I/O library may write a file successfully even if the application prepared the data incorrectly. Correctness must be assessed at the application level, where scientific transformations occur. We examine this in VPIC-Kokkos, a performance-portable Kokkos version of the Vector Particle-in-Cell (VPIC) application [1]. We reorganize output data by modifying ghost-cell removal, array layouts, and particle representations. Because these changes affect downstream analysis, we must show that exported data preserves the intended values and scientific interpretation across representation changes.

Prior VPIC work evaluated I/O performance in simulations and kernels [2], [3] and scientific accuracy under memory optimizations [4]. More broadly, scientific-software testing uses varying notions of correctness and standards of evidence [5]. We examine how complementary validation methods can provide evidence for application-level I/O optimizations and clarify what each method does and does not establish.

II. VALIDATION APPROACHES

A test oracle supplies an expectation against which to judge an output. We use a *deterministic export oracle* for prescribed output values and a *scientific oracle* for expected physical behavior.

For the deterministic export oracle, we initialize small VPIC problems with prescribed particle and field patterns and export them at step zero. Python verifiers check selected properties of binary and Hierarchical Data Format version 5 (HDF5) output with one or four processes. They compare prescribed values and output modes using fixed absolute tolerances. The HDF5 verifier compares separately sorted normalized momentum components. VPIC includes the deterministic test in its CMake test suite for repeatable development execution.

For the scientific oracle, we classify exported particles by dominant momentum direction and compute directional energy spectra. A scientist compares the spectra with expected behavior. We expect isotropy in a uniform thermal-plasma control with no laser and different forward and backward spectra along the laser axis in a laser-driven case. This check remains separate from the automated deterministic test.

III. OBSERVATIONS AND THEIR SCOPE

Consider momentum pairs (u_x, u_y) : $(1, 10)$ and $(2, 20)$. If we swap only the second components, we get $(1, 20)$ and $(2, 10)$. The separately sorted components are identical, so such checks pass even though the particle records differ. Because energy depends on each pair's components, this swap can change energies. This highlights a limitation of the deterministic export oracle: comparing separately sorted components cannot verify within-particle associations. It clarifies the comparison's scope, not a VPIC export fault. The thermal-plasma control spectra are isotropic as expected. In the laser-driven case, forward and backward spectra are unexpectedly similar, contradicting the expected directional behavior. The oracle flags this discrepancy, but the observation alone cannot determine whether it stems from the simulation, export, or analysis. These observations show why each oracle's scope matters. Matching expected spectra doesn't verify every exported record, and spot-checking exports doesn't validate the physical model or the simulation's full time evolution. The scientist's assessment is useful evidence, even though the discrepancy's source remains unresolved.

IV. COMMUNITY DISCUSSION

What correctness evidence should accompany an application I/O optimization? Our preliminary results suggest reporting the expected properties, the checks and tolerances used, and what remains untested. For scientific oracles, this includes expected physical behavior and the scientist's assessment; for deterministic export oracles, prescribed inputs and repeatable assertions. Making this scope explicit clarifies what a passing check establishes and what a disagreement can reveal. Reporting this evidence alongside performance results helps developers and scientists assess what an I/O optimization preserves.

Future work will strengthen record-level checks and tolerance thresholds, potentially enabling fuzzing to explore cases beyond prescribed tests [6].

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